NEIU Parking Management System

Vision and Scope

Project Name: NEIU Parking Management System

1. Vision

The NEIU Parking Management System provides a centralized, reliable, and efficient platform for handling parking data at Northeastern Illinois University. It simplifies the registration process for students, staff, vehicles, parking permits, lots, and events, replacing manual records with a digital system that enhances operational efficiency, data accuracy, traceability, and reporting. Overall, the system supports NEIU's goal of maintaining a secure, organized, and accessible parking infrastructure on campus, while also reducing administrative workload, lowering errors, and improving decision-making.

2. Business Objectives

- **Keep accurate and current records of students, employees, vehicles, and permits.**
- **Make issuing and managing parking permits easier.**
- **Offer insights into parking usage through event tracking.**
- **Aid enforcement, compliance, and resource planning.**
- **Cut down on manual data entry, paper forms, and repetitive admin tasks.**
- **Build a foundation for future upgrades like online access, automated enforcement, and analytics.**

3. Scope

3.1 In-Scope Functionality

The application will provide functionality for:

A. Student and Employee Management

- **Add, view, update, and delete student records.**
- **Add, view, update, and delete employee records.**

B. Vehicle Management

- **Register vehicles linked to students or employees.**
- **Associate vehicles with zero or one student, or zero or one employee.**
- **Remove or update existing vehicles.**
- **Enforce referential integrity.**

C. Permit Management

- **Define permit types (e.g., Gold, Blue, Yellow).**
- **Issue parking permits with valid dates.**
- **Associate permits with vehicles.**
- **Track multiple permits per vehicle over time.**

D. Parking Lot & Space Management

- **Define parking lots and assign unique spaces.**
- **Manage space identification and availability (structural, not real-time).**

E. Parking Event Tracking

- **Record vehicle entries and exits.**
- **Associate events to lots and permits.**

F. Data Views and Reporting

- **Display student-vehicle table.**
- **Display employee-vehicle table.**
- **Display permit-vehicle status and validity.**

3.2 Out-of-Scope Functionality

The current version does not include:

- **Real-time sensor-based parking availability.**
- **Visualization dashboards or analytics.**
- **Automated citation or enforcement workflows.**
- **Mobile or web interface.**
- **User authentication or role-based access control.**
- **Payment processing.**
- **Integration with university systems.**

These may be addressed in future releases.

4. Stakeholders

- Primary Users: NEIU Parking Officer / Admin
- Indirect Beneficiaries: Students and employees (due to better service)

5. Proposed System Environment

- Java console application
- MySQL relational database
- JDBC for connectivity
- Structured database schema with referential integrity
- SQL views for consolidated data representation

6. Constraints and Assumptions

Constraints:

- System depends on an active MySQL instance.
- Data must conform to defined relationships and constraints.
- Users interact through a text-based menu system.

Assumptions:

- Users have basic technical literacy.
- Parking data is updated regularly and accurately.
- Database will grow over time and require maintenance.

7. Success Criteria

The project will be considered successful if it:

- **Accurately stores and retrieves parking-related data.**
- **Maintains referential integrity and prevents inconsistent entries.**
- **Reduces time and effort required to manage parking records.**
- **Provides clear, readable output to end-users.**
- **Supports future enhancements without redesigning the core database.**

8. Future Vision

The application is intended as a scalable foundation for future innovation, including:

- **A web or mobile front-end**
- **Real-time parking availability tracking**
- **Digital permit assignment integrated with student/employee accounts**
- **Reporting dashboards and analytics**
- **Integration with security and access-control systems**
- **Automated violation detection and alerts**

These developments would transform the NEIU Parking Management System from a record-keeping system into a comprehensive smart-campus parking management platform.

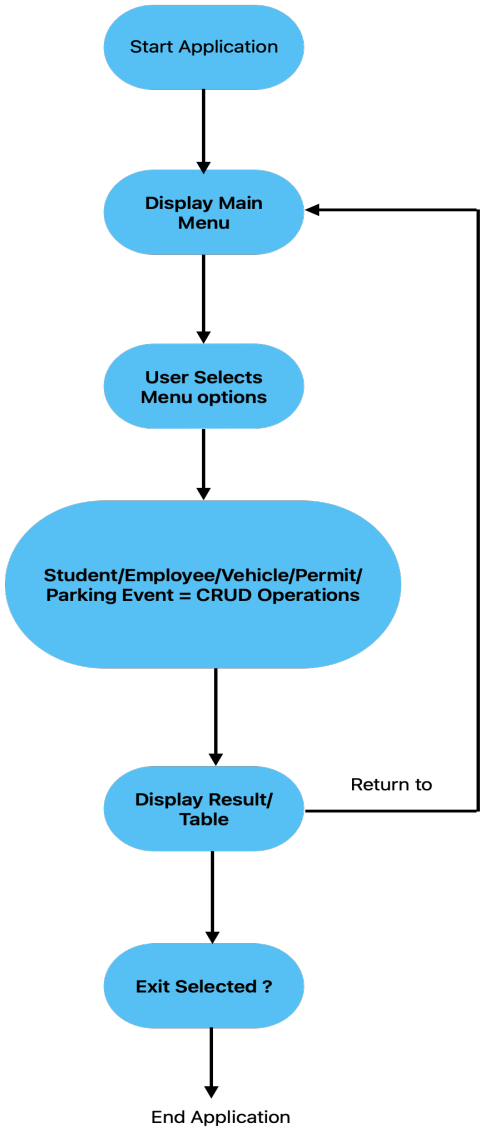
UC-1 TABLE VIEW

ID and Name:	NEIU Parking Management System		
Created By:	***** **********	**Date Created:**	**/**/****
Primary Actor:	Parking officer / Admin	**Secondary Actors:**	Database (MySQL) NEIU Students and Employee (not directly but as data records)
Description:	This use case describes how a user interacts with the NEIU Parking Lot Database Management System to manage parking-related information. Using a console-based menu, the user can add, view, update, and delete records for students, employees, vehicles, parking permits, parking lots, parking spaces, and parking events. The system validates input, enforces database constraints, and provides feedback after each operation.		
Trigger:	The admin starts the application and reaches the Main Menu.		
Preconditions:	The system is running and connected to the database. Student or Employee must exist in the system if the vehicle will be associated with that person. Also, if a student/employee is not yet in DB, they can be added via the application.		
Postconditions:	A new vehicle is stored in the DB. The data is available for future queries, permit assignments, and CRUD operations.		
Normal Flow:	1. User starts the Parking Management System application. 2. System displays the Main Menu with options, such as: Manage Student Manage Employee Manage Vehicles Manage Permits/Permit Types Manage Parking Lots/Spaces Manage Parking Events 3. User selects one of the managements options. 4. System displays a sub menu for the chosen entity, such as: Add(Create) View(Read) Update Delete Return to previous Menu		

	5. User Selects an operation. 6. System prompts the user for the required input. 7. User enters the requested data. 8. System validates the input. 9. System performs the requested database operation via JDBC. Executes SQL INSERT/SELECT/UPDATE/DELETE 10. System processes the result and displays the data via a formatted table. Upon failure, the system will display the error message to the user. 11. System returns the user to the relevant sub menu or main menu. 12. User may repeat steps 3-11 for additional operations. 13. When done, user can EXIT the application from the Main Menu. 14. System terminates the application.
Alternative Flows:	1. Invalid Menu Option User inputs a number that doesn't match to the corresponding menu option System displays "Invalid choice" 2. No records found User chooses to view a record that doesn't exist. System displays a message such as " No record found!" and returns to the menu.
Exceptions:	1. Invalid input format. User enters invalid data. System displays error and re-prompts for correct input. 2. Database connection error System cannot insert data due to SQL error. System informs users of system error and returns back to main menu. 3. Foreign Key Violation 4. Duplicate Primary Key
Priority:	High. Essential System Functionality.
Frequency:	Frequent. Expected multiple times per day by the admin or parking office.
Business Rules:	License plate must be unique. Owner can be either a current student or NEIU employee, even though, they can be added before adding a vehicle. Certain deletes may be restricted or cascade. Data must be kept consistent with defined foreign keys and ON DELETE rules (CASCADE / SET NULL)

Other Information:	This use case is part of the core CRUD process for the Parking Management System.
Assumptions:	User has basic computer skills. User understands NEIU parking business concepts. Database contains valid student and employee data. System validates data before insertion.

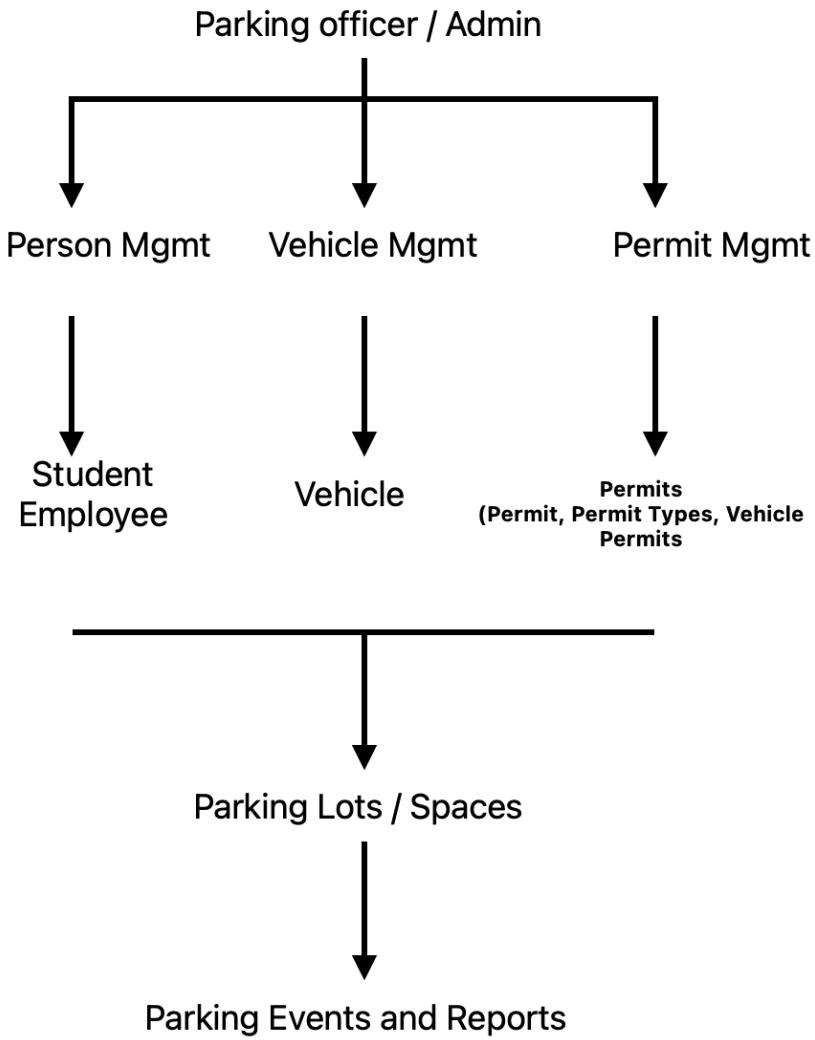
UC-2. Activity Diagram



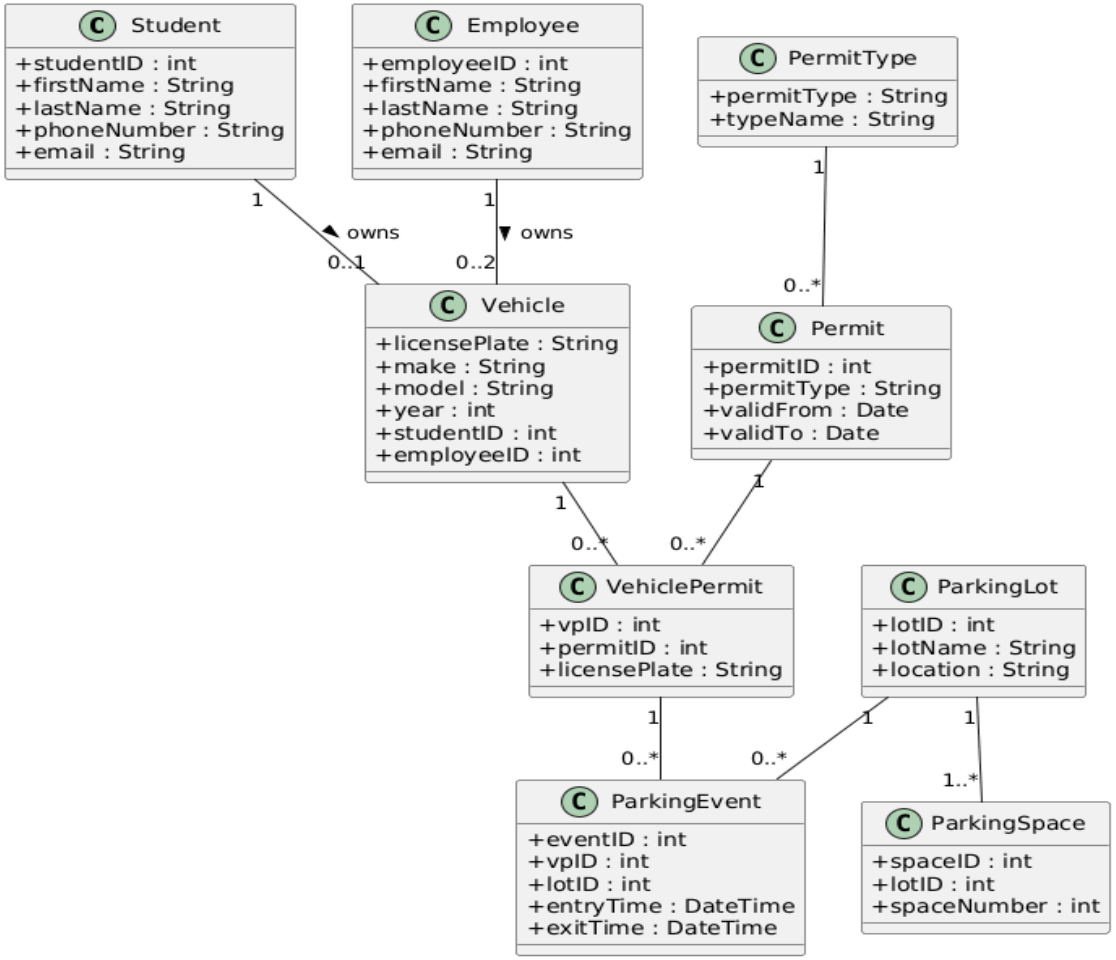
NEIU Parking Management System

1. The system displays the main Menu.
2. User selects an operation.
3. Application performs CRUD logic via JDBC
4. The database returns data or an error
5. The system displays output
6. User repeats or exits

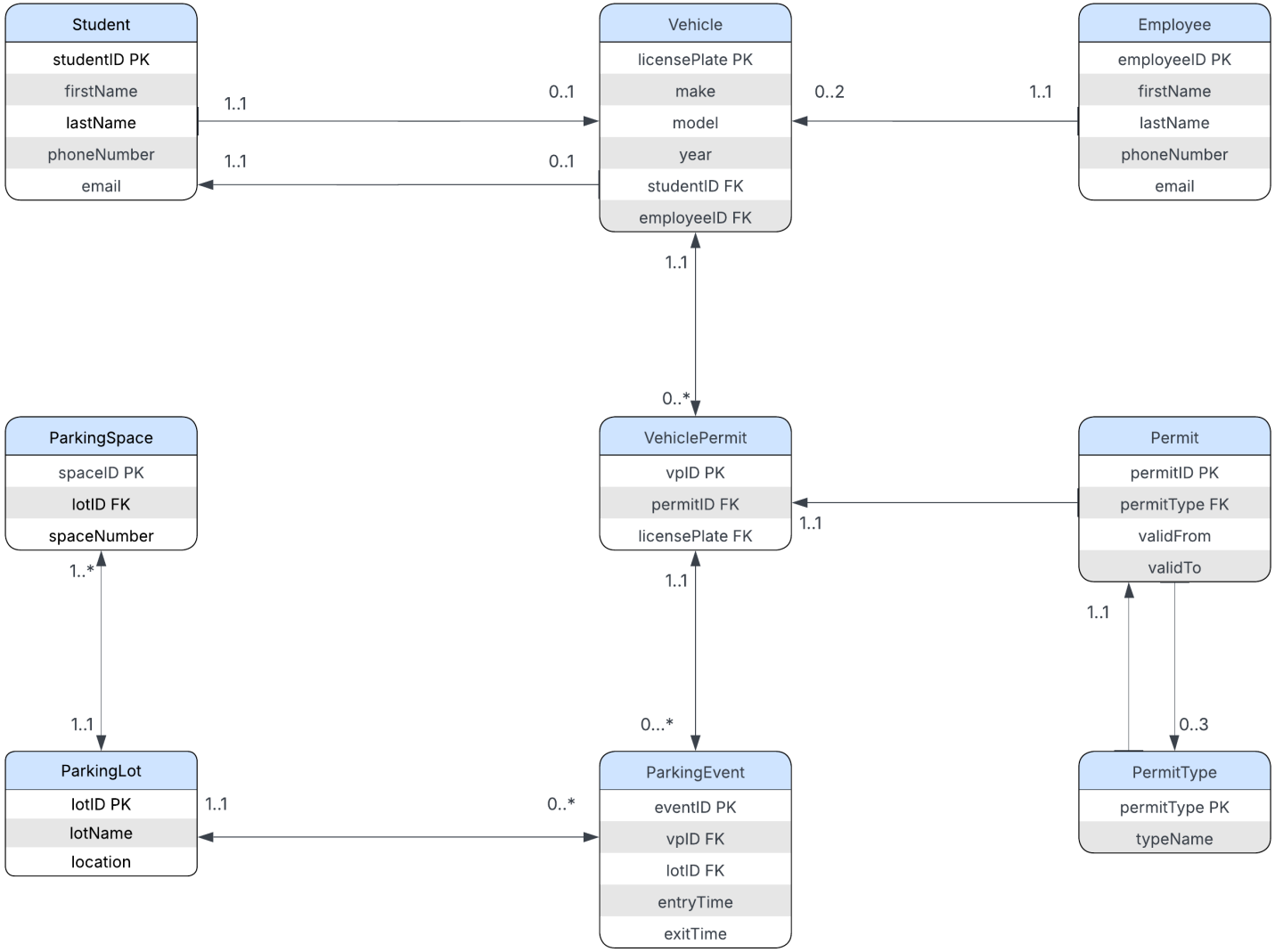
UC – 3. Data Flow Diagram



UC – 4 UML Class Diagram



UC – 5. ER Diagram



The NEIU Parking Management System tracks information about students, employees, vehicles, parking permits, and parking activities in parking lots. Students can register up to one car, and employees can register up to two vehicles. Each vehicle can have one active permit at a time but may be assigned multiple permits over time due to renewal or expiration. A permit is linked to a specific permit type, such as Gold, Blue, or Yellow, and can be assigned to multiple vehicle permits over time. Each vehicle permit logs parking events, including entry and exit times when a vehicle enters a parking lot. Parking lots are composed of multiple parking spaces, and each parking event is linked to the lot where it occurred.

Referential integrity constraints with CASCADE action are established for the VehiclePermit Table, ParkingEvent Table, and ParkingSpace Table. When a permit is deleted, the associated vehicle permit is also removed from the vehicle, and similarly, deleting an entire parking lot results in the removal of all parking spaces from the database. If any of these actions occur, the related data in the ParkingEvent is also deleted.

The PermitType table is designed as a reference table because those permit types rarely change and are intended for long-term use. The Permit Table will be updated each semester with new validFrom and validTo dates, and consequently, a permit will be assigned to a vehicle (vpID).

There are three views implemented so far: Student Vehicles, Employee Vehicles, and Vehicle Permit Info. The application displays the denormalized Vehicle Permit Info view to users instead of directly exposing the normalized VehiclePermit table, since the view provides more meaningful information.

The database is fully normalized, reducing redundancy, preventing anomalies, and enforcing logical constraints.